Detecting Pump-and-Dump in Small-Cap Ethereum Tokens

AC297r — Mid-Semester Presentation

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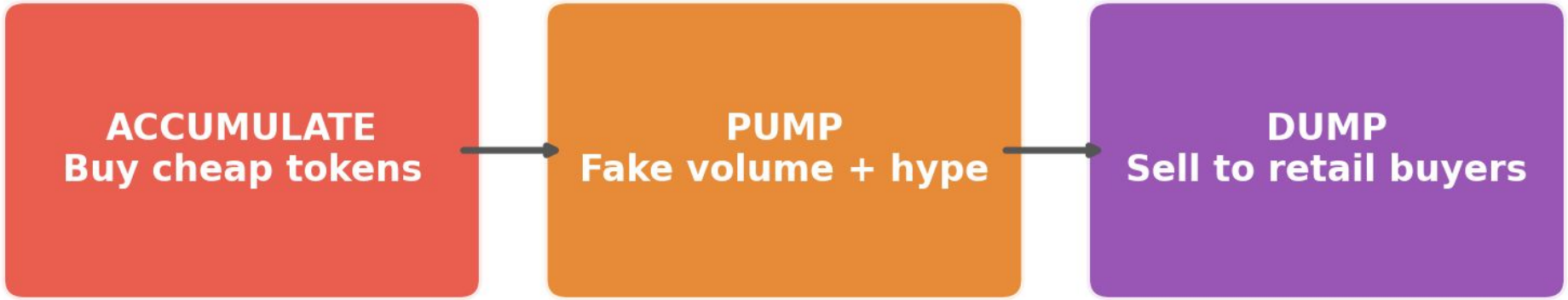
Team & Industry Partners

Our Team		Industry Partners (Taiwan)	
Gillian Schriever	Project management, analytics	**@Lumberg**	Crypto investor & entrepreneur
Jiahui (Cecilia) Cai	ML, causal modeling, exchange flow	**@MiLLi3e**	Ethereum specialist, cofounder of Trueo
Zhilin Chen	LLMs, investment banking analytics	**@uberjake**	Dune Analytics expert (MIT)
Jinghan Huang	Economics, wallet concentration & wash trading	**@eevoy**	Prediction markets researcher (Harvard)

Team strengths: data engineering, statistical modeling, financial analytics, project management

Problem & Solution



Why This Matters

- **$5.6B** lost to crypto fraud in 2023 (FBI Report)
- Small-cap tokens are most vulnerable: low liquidity, few holders, easy to manipulate
- Price data alone cannot distinguish manipulation from organic volatility

Problem	Solution
No reliable way to detect pump-and-dump using only market price data.	Use on-chain blockchain data as features. Output: fraud probability + confidence interval.

Data: Dune Analytics — Ethereum blockchain (token transfers, Decentralized Exchange trades, exchange wallets)

Project Goals & Success Criteria

G1
Feature Extraction

5 on-chain feature categories for 5,000+ Ethereum tokens

G2
Fraud Probability Model

Probabilistic score + 90% confidence interval per token

G3
Signal Interpretation

Which on-chain signals are most predictive? (SHAP*)

Success Criteria

Goal	Success Criterion
G1	Scalable pipeline that produces complete feature vectors for **5,000 Ethereum tokens**
G2	Each token receives a fraud probability estimate with a **90% confidence interval**, validated against **20 known cases**
G3	Blockchain-derived features demonstrate **at least 5% measurable improvement** over price and volume data alone

**SHAP (SHapley Additive exPlanations) assigns a contribution value to each feature showing how much it pushed the fraud prediction up or down.*

Methodology

Two approaches under consideration — same feature set, different assumptions

A: Supervised Classification (primary)

Label: auto-label + partner-verified cases

Price spike ≥5x + crash ≥80% = suspected P&D

Looking for verified fraud instances

Model: XGBoost / Random Forest

Output: fraud probability + 90% CI

Quantitative score, not qualitative buckets

Provide full distribution when possible

B: Unsupervised Anomaly Detection (backup)

No labels needed:

Treats fraud tokens as outliers

Methods:

Isolation Forest

DBSCAN Clustering

PCA + Mahalanobis Distance

Validate: cross-check with known cases

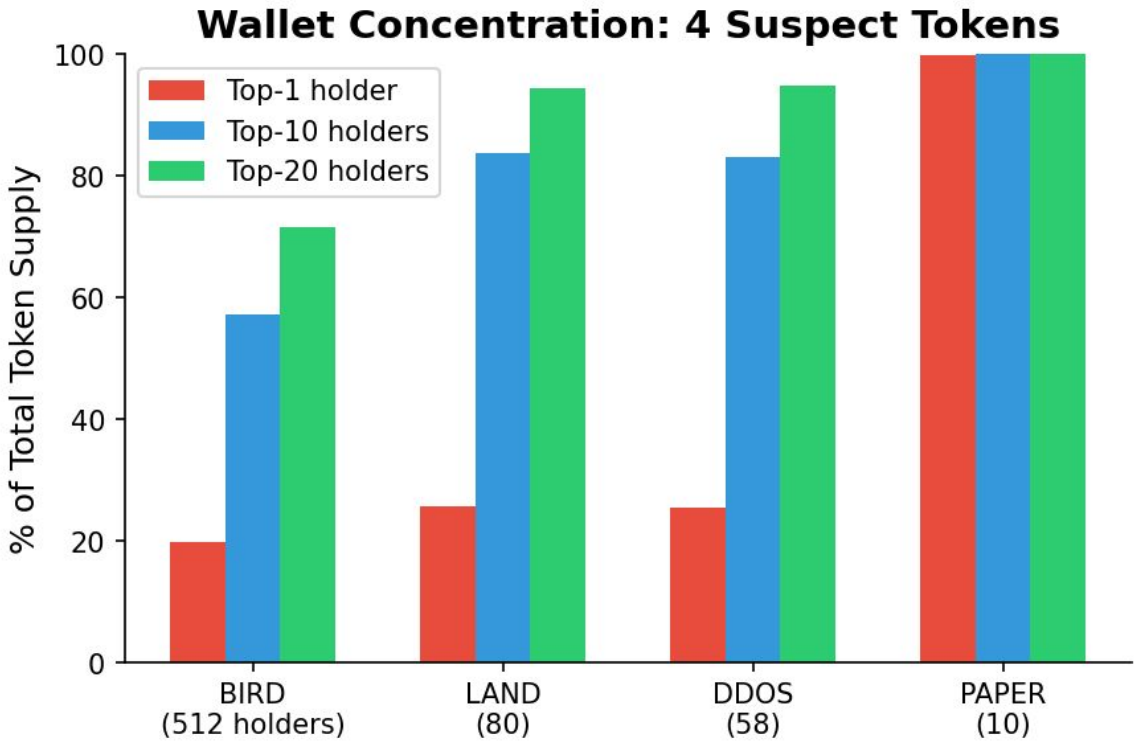
Shared Feature Set (5 categories)

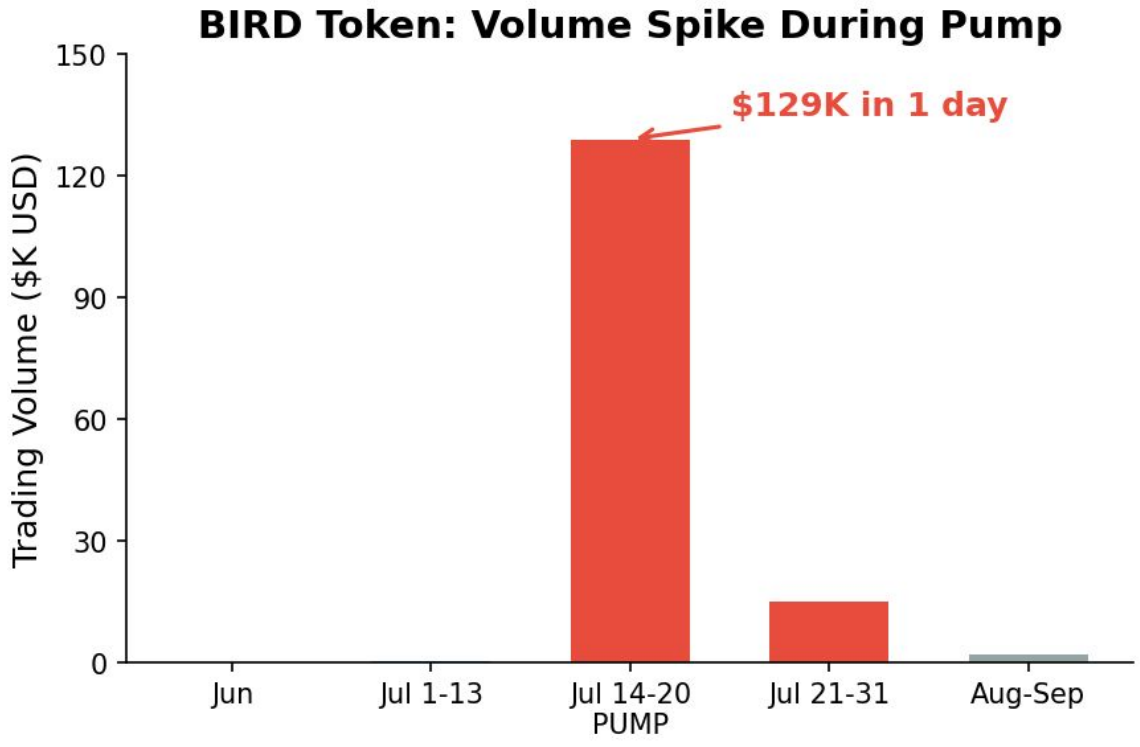
Wallet Concentration → Volume Surge → Transfer Graph → Wallet Behavior → Exchange Flow

Decision at Week 10: if labels are reliable → A. If too noisy → B. Both share features — no duplicated work.

Current Analysis

✓ 8 SQL queries ✓ 4 tokens analyzed ✓ 5 blockchain features extracted





BIRD: single owner → wash traded 21% of volume → dumped during $129K spike

Plan

Week	Task	Goal	Owner
7	Mid-semester presentation	—	All
8	Screen & auto-label 5,000+ tokens; ask partners & search for labeled fraud cases	G1, G2	Jinghan, Zhilin
9	Feature extraction at scale (5 modules)	G1	Cecilia, Gillian
10	Data quality check; decide method A vs B	G1, G2	All
11	Train model; calibrate probability + CI	G2	Jinghan, Zhilin
12	SHAP* analysis; baseline comparison	G3	Cecilia, Gillian
13	Integration & sensitivity analysis	All	All
14	Final report & presentation	—	All

Key decision: Week 10 — label quality determines supervised (A) vs unsupervised (B)

**SHAP (SHapley Additive exPlanations) assigns a contribution value to each feature showing how much it pushed the fraud prediction up or down.*

Open Questions

Labeled Data	No ground-truth fraud labels exist	Ask partners or search for known P&D instances; even a small labeled set helps calibration
Thresholds	One threshold won't fit all tokens; scores may need normalization by subgroup	Normalize by token volume/holder count; sensitivity analysis on thresholds
Governance Tokens	Some tokens let majority holders change rules — a different fraud vector	Flag tokens with governance; explore as additional risk feature
Why High Concentration?	Top wallets may be minters or legitimate founders, not manipulators	Trace wallet origin (deployer? minter?); distinguish insiders from founders
Method Selection	Supervised vs unsupervised: TBD	Both use same features; decide at Week 10 based on data